

Computer Science & Information Technology

Computer Network - 2

DPP: 1

Transport & Application Layer

- Q1** Which of the following socket API function is/are executed by TCP server?
(A) bind() (B) connect()
(C) listen() (D) accept()
- Q2** Consider a TCP connection with end-to-end transmission bandwidth is 262140 bits/sec. What value of RTT in seconds (rounded off to the nearest integer) after which the TCP window scale option is needed?
- Q3** Consider 'window scale' option is opted during TCP connection establishment phase, the maximum possible window size with window scale option is $(2^{16} - 1) * 2^N$. What is the value of N?
- Q4** Identify which name server(s) does not belong to DNS hierarchy?
(A) Root Name Server
(B) Top Level Domain Name Server
(C) Real Name Server
(D) Authoritative Name Server
- Q5** Consider the resolution of the domain name "www.iitd.ac.in" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.
- Q6** Consider the resolution of the domain name "www.google.com" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. Let RTT denote the round trip time between DNS resolver and DNS servers in the DNS hierarchy. The number of RTTs required in completely resolving the domain name is _____.
- Q7** Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP. The web page contains base HTML file and four very small image objects. How many TCP connections required to load the web page completely?
- Q8** Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP (no any parallel TCP connections). The web page contains base HTML file and four very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?
- Q9** Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP that support parallel TCP connections. The web page contains base HTML file and four very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?
- Q10** Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP without pipeline. The web page contains base HTML file and four very small image objects. Assume base HTML file and all



four image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?

- Q11** Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP with pipeline. The web page contains base HTML file and four very small image objects. Assume base HTML file and all four image objects are placed in same web server. How many total number of RTTs (Round

Trip Time) it takes to load the web page completely ?

- Q12** Which is/are valid HTTP request method ?
(A) GET (B) PUT
(C) POST (D) HEAD
- Q13** Consider a web browser at client machine made a request for a web page to a web server. Identify the correct sequence in which the following protocols are used by the client host?
(A) HTTP, TCP, DNS (B) DNS, TCP, HTTP
(C) DNS, HTTP, TCP (D) TCP, DNS, HTTP



Answer Key

Q1 (A, C, D)

Q2 2~2

Q3 14~14

Q4 (C)

Q5 4~4

Q6 3~3

Q7 5~5

Q8 10~10

Q9 4~4

Q10 6~6

Q11 3~3

Q12 (A, B, C, D)

Q13 (B)



[Android App](#) | [iOS App](#) | [PW Website](#)

Hints & Solutions

Q1 Text Solution:

connect() → TCP client

Q2 Text Solution:

Without window scale option

TCP receiver can advertise largest window size = $(2^{16} - 1)$ byte

$$\begin{aligned} \text{Max RTT (without window scale)} &= \frac{(2^{16}-1) \text{ bytes}}{\text{Bandwidth}} \\ &= \frac{(2^{16}-1) \times 2^3 \text{ bits}}{262140 \text{ bits/sec.}} = 2 \text{ sec.} \end{aligned}$$

Q3 Text Solution:

Maximum possible window(with window scaling) = $(2^{16} - 1) \times \text{Scale Factor}$

Scale Factor = $2^{\text{Shift_Count}}$

In TCP maximum value for shift count can be 14.

Q4 Text Solution:

Real Name Server

Q5 Text Solution:

Domain Name : "www.iitd.ac.in"

Mode : Iterative Query

DNS Resolver → 4-tier DNS hierarchy

Q6 Text Solution:

Domain name : "www.google.com"

Mode : Iterative Query

DNS Resolver → 3-tier DNS hierarchy

Q7 Text Solution:

Non persistent HTTP:

Web page = [Base HTML file + 4 Small image objects]

No. of TCP Connection = $(1 + 4) = 5$

Q8 Text Solution:

Non-persistent HTTP:

Web page = [Base HTML File + 4 Small image objects]

Total time = 2 RTT + $4 \times 2 \text{ RTT} = 10 \text{ RTT}$

Q9 Text Solution:

Non-persistent HTTP:

Web page = Base HTML File + 4 Very small image objects (Using parallel TCP Conn.)

Total time = 2 RTT + 2 RTT = 4 RTT

Q10 Text Solution:

Persistent HTTP:

1 RTT → for TCP connection

1 RTT → for Base HTML File

1 RTT → first object request

...

1 RTT → fourth object request

Total = 6 RTT

Q11 Text Solution:

Persistent HTTP:

1 RTT → for TCP connection

1 RTT → for Base HTML file

1 RTT → for all four objects request in pipeline

Total = 3 RTT

Q12 Text Solution:

All four options are correct.

Q13 Text Solution:

(1) DNS:

(i) DNS Query

(ii) DNS Reply

(2) TCP:

(i) TCP Connection Request

(ii) TCP Accept Response

(3) HTTP:

(i) HTTP Request Message

(ii) HTTP Response Message

